

Gray-Scale

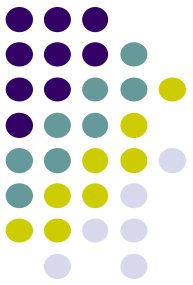
Morphological Image Processing

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Faculty of Computer Engineering

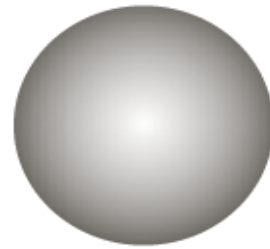
University of Isfahan



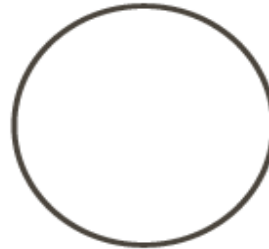
Gray-Scale Morphology

$f(x, y)$: gray-scale image

$b(x, y)$: structuring element



Nonflat SE



Flat SE



Intensity profile



Intensity profile

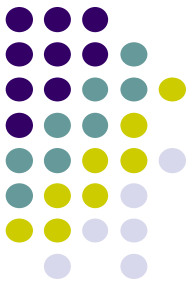


FIGURE 9.34

Nonflat and flat structuring elements, and corresponding horizontal intensity profiles through their center. All examples in this section are based on flat SEs.

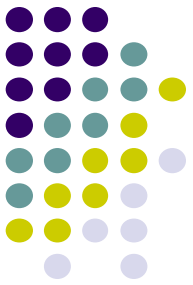


Gray-Scale Morphology: Erosion and Dilation by Flat Structuring



$$[f \ominus b](x, y) = \min_{(s,t) \in b} \{ f(x + s, y + t) \}$$

$$[f \oplus b](x, y) = \max_{(s,t) \in b} \{ f(x - s, y - t) \}$$



$$(f \oplus b)(s) = \max \{f(s-x) + b(x) \mid (s-x) \in D_f; x \in D_b\}$$

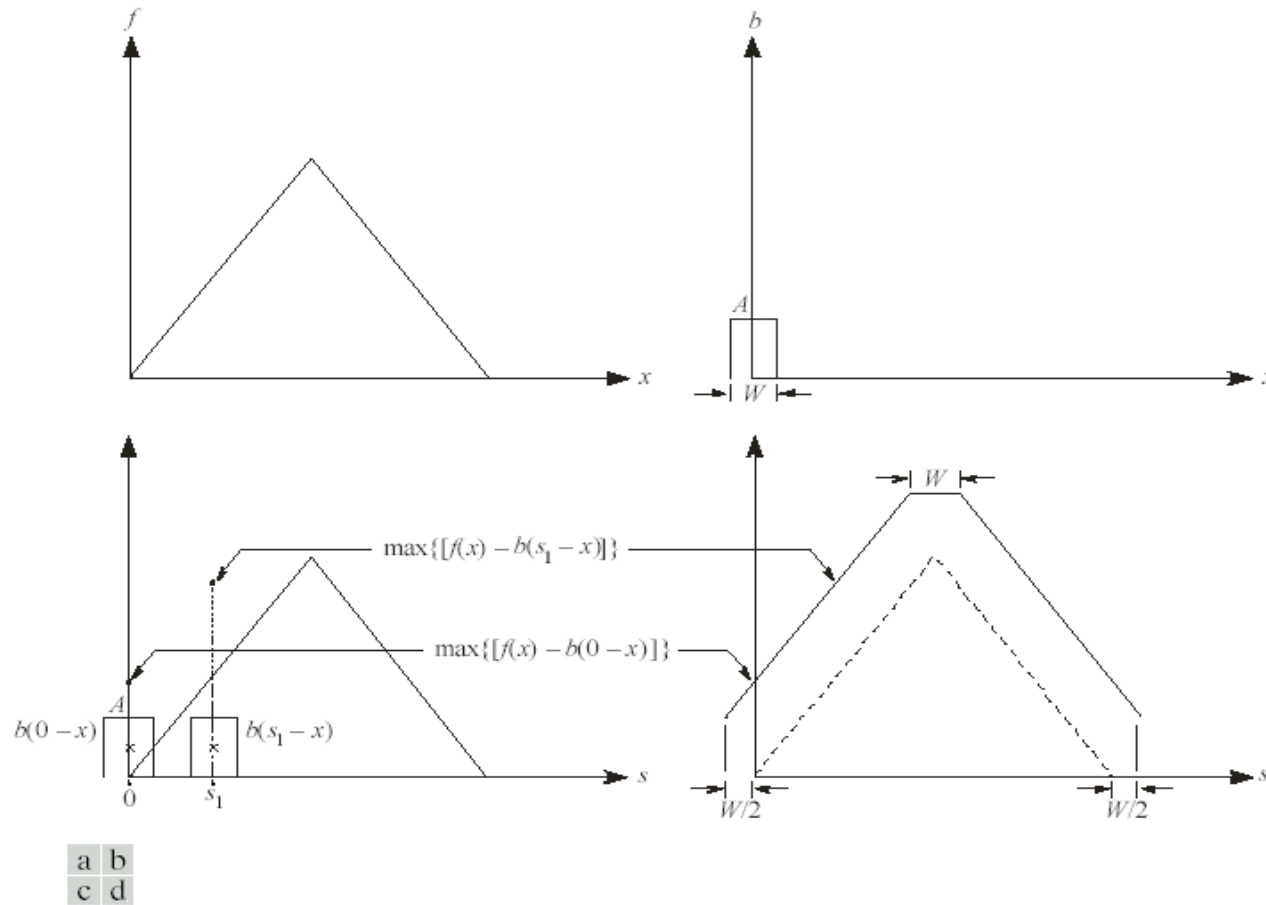
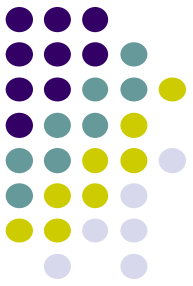


FIGURE 9.27 (a) A simple function. (b) Structuring element of height A . (c) Result of dilation for various positions of sliding b past f . (d) Complete result of dilation (shown solid).



G-Level Dilation General effect

- If all the values of the structuring element are positive, the output image tends to be **brighter than the input.**
- Dark details either are reduced or eliminated, depending on how their values and shapes relate to the structuring element used for dilation.

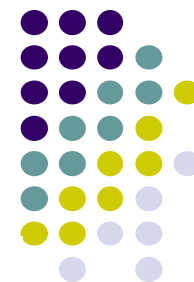
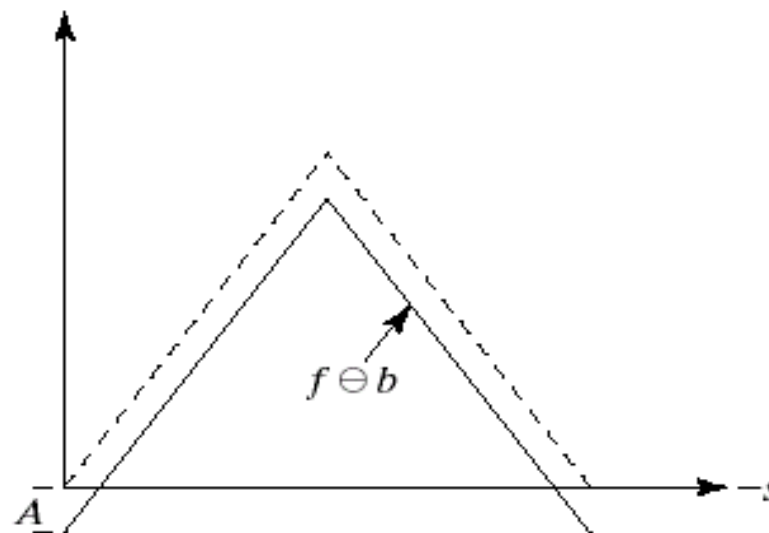
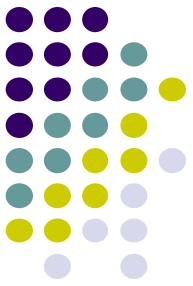


FIGURE 9.28

Erosion of the function shown in Fig. 9.27(a) by the structuring element shown in Fig. 9.27(b).



$$(f \ominus b)(s) = \min \{ f(s+x) - b(x) \mid (s+x) \in D_f; x \in D_b \}$$

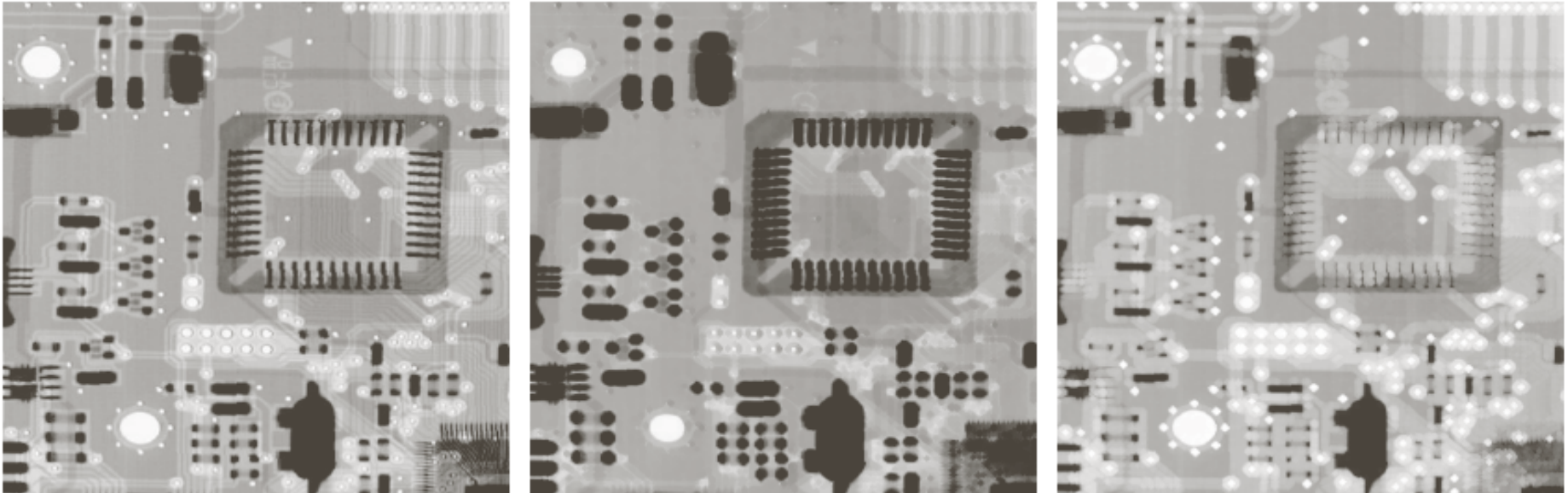
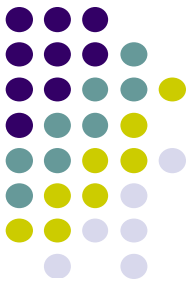


G-Level Erosion General effect

- If all the elements of the structuring element are positive, the output image tends to **be darker than the input.**
- The effect of **bright details** in the input image that are smaller in area than the structuring element is reduced, with the degree of reduction being **determined** by the gray-level values surrounding the bright detail and by the shape and amplitude values of the structuring element itself.



Example 1

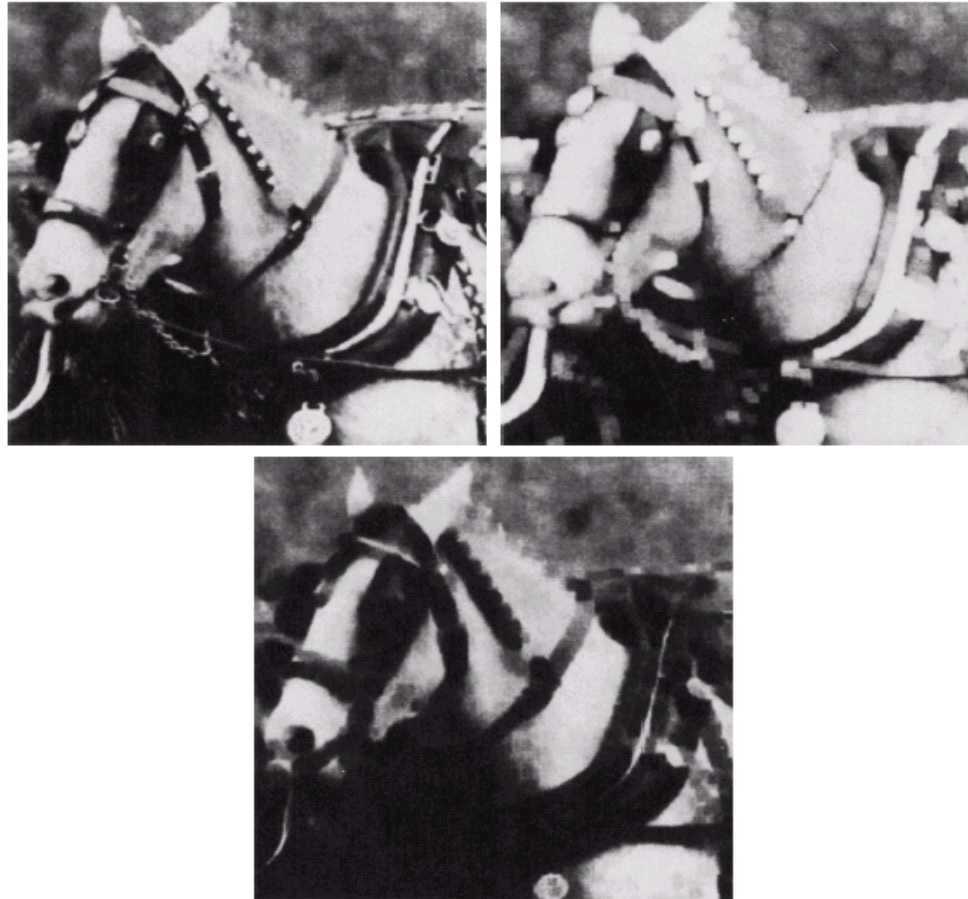
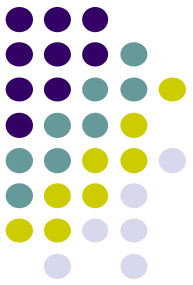


a b c

FIGURE 9.35 (a) A gray-scale X-ray image of size 448×425 pixels. (b) Erosion using a flat disk SE with a radius of two pixels. (c) Dilation using the same SE. (Original image courtesy of Lixi, Inc.)



Example 2



a b
c

FIGURE 9.29

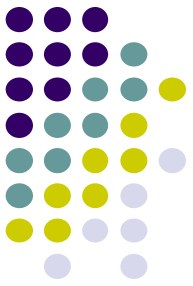
(a) Original image. (b) Result of dilation.

(c) Result of erosion.

(Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

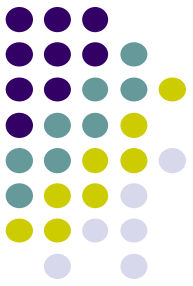


Gray-Scale Morphology: Erosion and Dilation by Nonflat Structuring



$$\left[f \ominus b_N \right](x, y) = \min_{(s,t) \in b} \left\{ f(x + s, y + t) - b_N(s, t) \right\}$$

$$\left[f \oplus b_N \right](x, y) = \max_{(s,t) \in b} \left\{ f(x - s, y - t) + b_N(s, t) \right\}$$



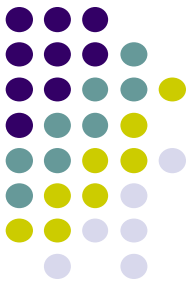
Duality: Erosion and Dilation

$$[f \ominus b]^c(x, y) = \left(f^c \oplus \hat{b} \right)(x, y)$$

where $f^c = -f(x, y)$ and $\hat{b} = b(-x, -y)$

$$[f \ominus b]^c = \left(f^c \oplus \hat{b} \right)$$

$$(f \oplus b)^c = (f^c \ominus \hat{b})$$



Opening and Closing

$$f \circ b = (f \ominus b) \oplus b$$

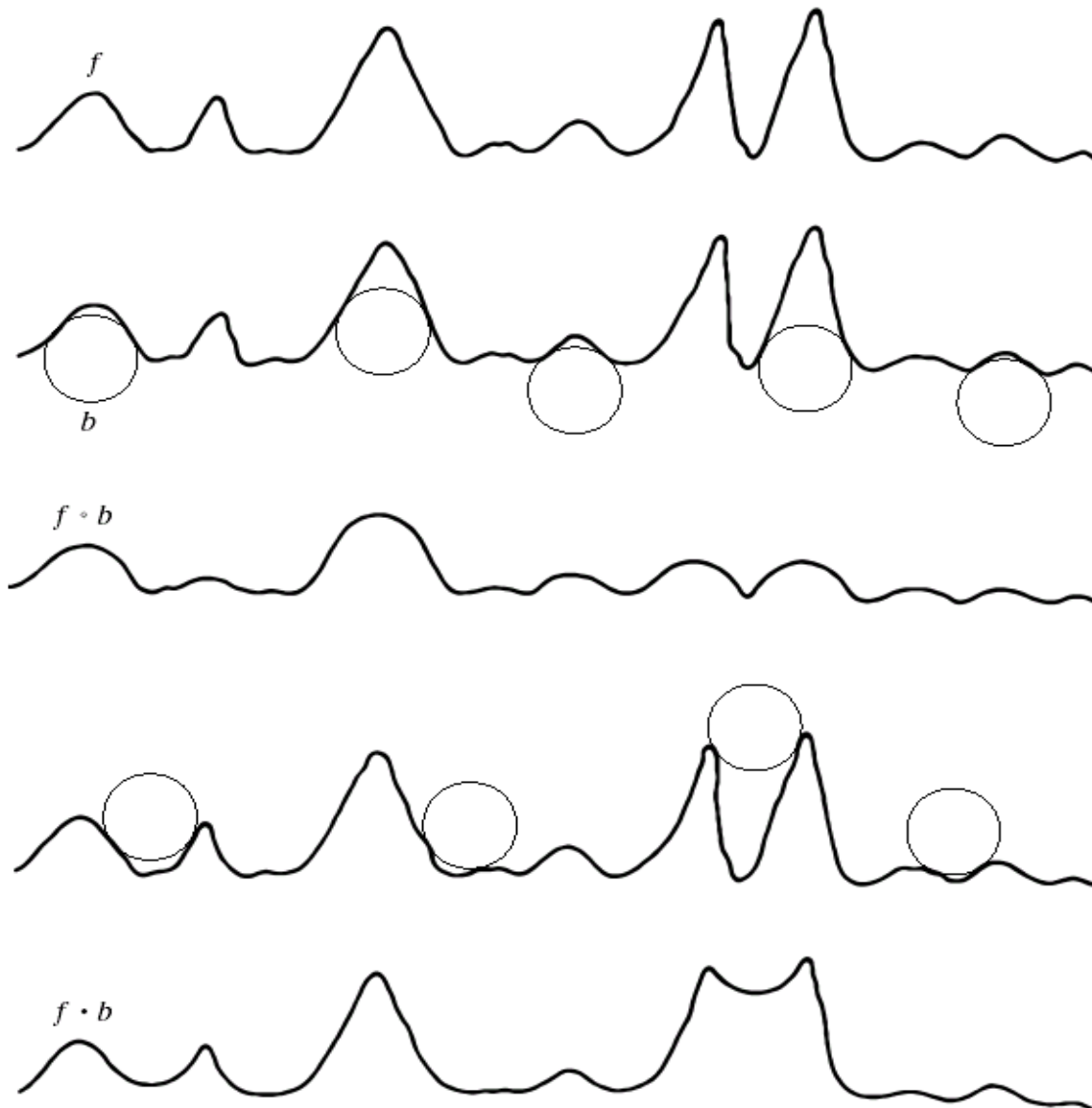
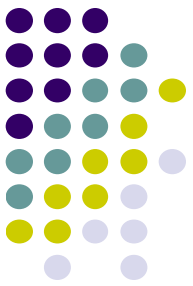
$$f \bullet b = (f \oplus b) \ominus b$$

$$(f \sqcap b)^c = f^c \circ \hat{b} = -f \circ \hat{b}$$

$$(f \circ b)^c = f^c \sqcap \hat{b} = -f \sqcap \hat{b}$$



Opening and Closing in 1 dim.



a
b
c
d
e

FIGURE 9.30

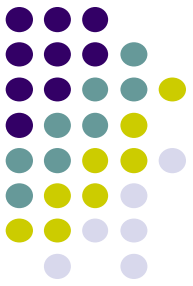
(a) A gray-scale scan line.

(b) Positions of rolling ball for opening.

(c) Result of opening.

(d) Positions of rolling ball for closing.

(e) Result of closing.



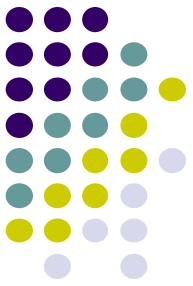
Properties of Gray-scale Opening

$$(a) \quad f \circ b \leftarrow \sqcup f$$

$$(b) \quad \text{if } f_1 \leftarrow \sqcup f_2, \text{ then } (f_1 \circ b) \leftarrow \sqcup (f_2 \circ b)$$

$$(c) \quad (f \circ b) \circ b = f \circ b$$

where $e \leftarrow \sqcup r$ denotes e is a subset of r and also $e(x, y) \leq r(x, y)$.



Properties of Gray-scale Closing

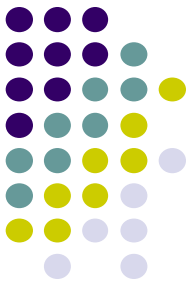
$$(i) \quad f \leftarrow \sqcup (f \bullet b)$$

$$(ii) \quad \text{if } f_1 \leftarrow \sqcup f_2, \text{ then } (f_1 \bullet b) \leftarrow \sqcup (f_2 \bullet b)$$

$$(iii) \quad (f \bullet b) \bullet b = (f \bullet b)$$



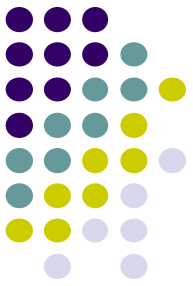
Opening General Effects



- **Opening is used to remove small light details, while leaving the overall gray levels and larger bright features relatively undisturbed.**
- **The initial erosion removes the details, but it also darkens the image.**
- **The subsequent dilation again increases the overall intensity of the image without reintroducing the details totally removed by erosion.**



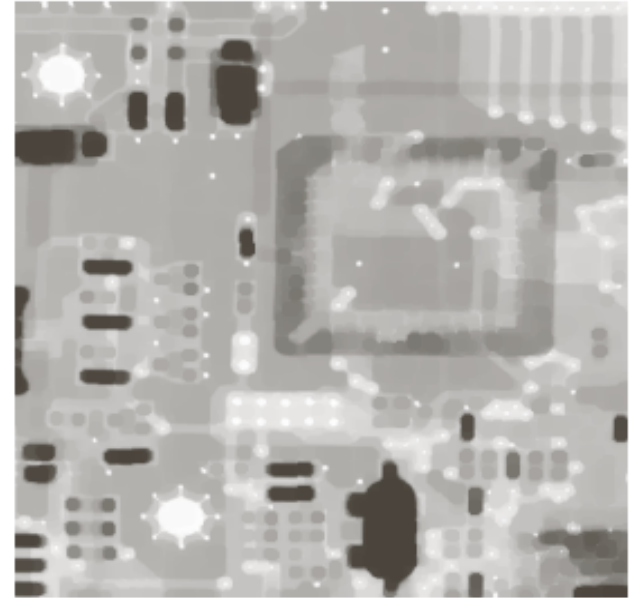
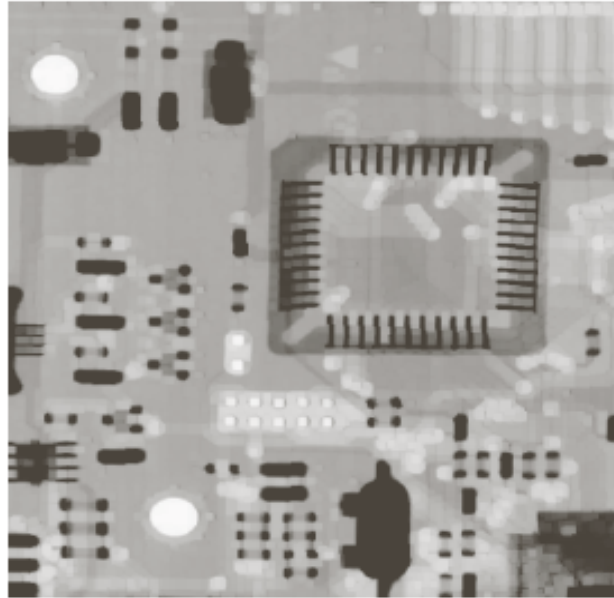
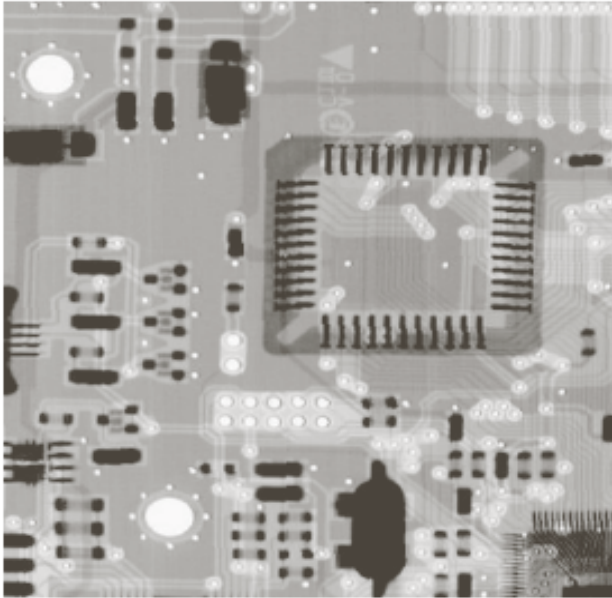
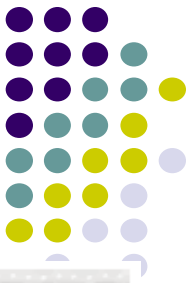
Closing General Effects



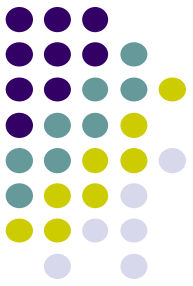
- **Closing is used to remove small dark details, while leaving bright features relatively undisturbed.**
- **The initial dilation removes the dark details and brightens the image**
- **The subsequent erosion darkens the image without reintroducing the details totally removed by dilation**



Example 1



a b c

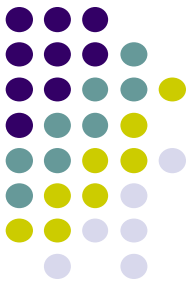


a b

FIGURE 9.31 (a) Opening and (b) closing of Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

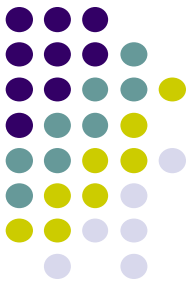
Opening: Decreased size of small bright details. No changes to dark region

Closing: Decreased size of small dark details. No changes to bright region



Morphological Smoothing

- **Opening suppresses bright details smaller than the specified SE, and closing suppresses dark details.**
- **Opening and closing are used often in combination as morphological filters for image smoothing and noise removal.**



Example 1



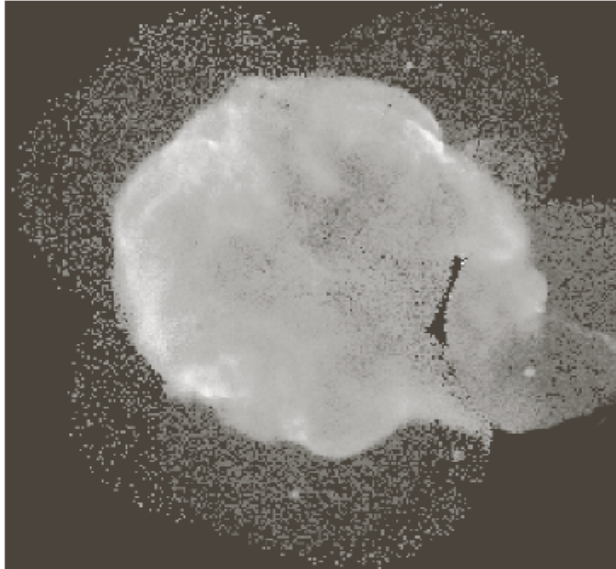
FIGURE 9.32 Morphological smoothing of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Smoothing

$$g = ((f \circ b) \bullet b)$$

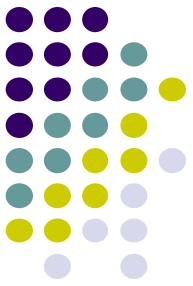


Example 2



a	b
c	d

FIGURE 9.38
(a) 566×566 image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b)–(d) Results of performing opening and closing sequences on the original image with disk structuring elements of radii, 1, 3, and 5, respectively. (Original image courtesy of NASA.)

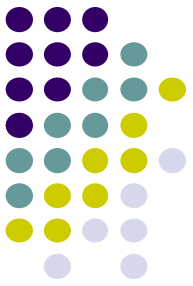


Morphological Gradient

- Dilation and erosion can be used in combination with image subtraction to obtain the morphological gradient of an image, denoted by g ,

$$g = (f \oplus b) - (f \ominus b)$$

- The edges are enhanced and the contribution of the homogeneous areas are suppressed, thus producing a “derivative-like” (gradient) effect.



Example 1

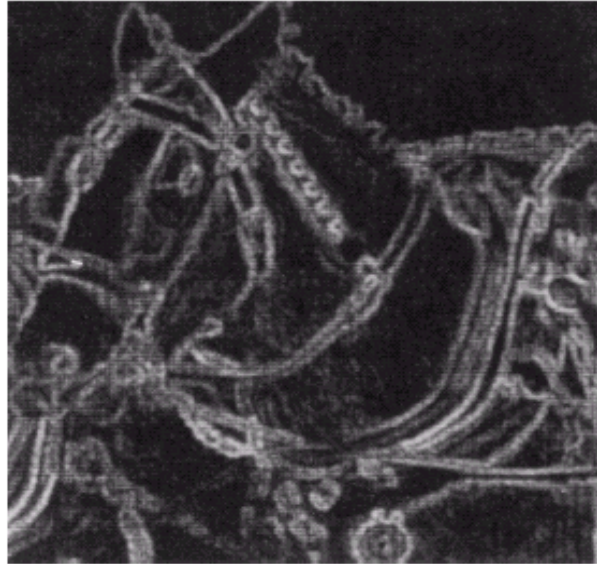


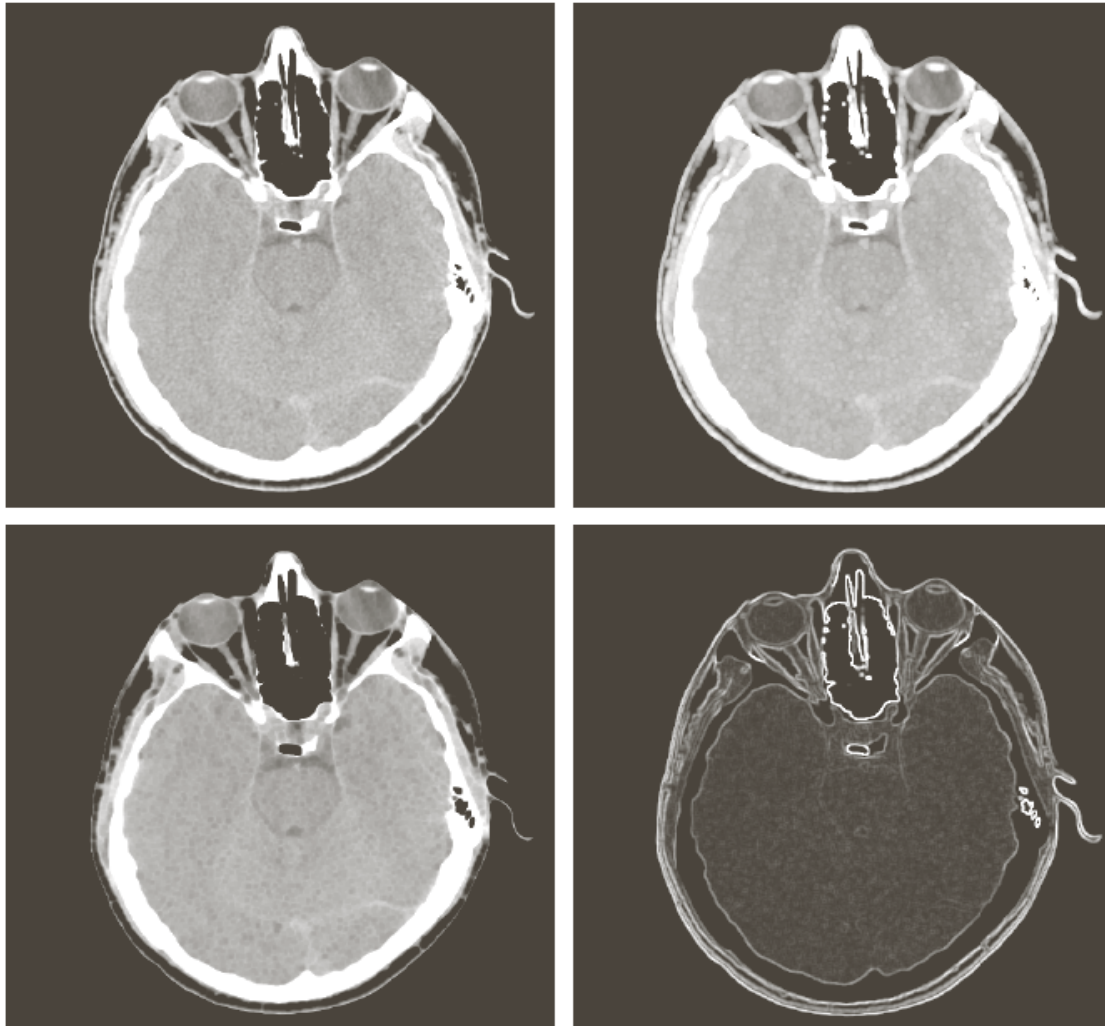
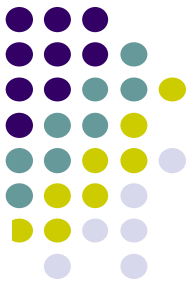
FIGURE 9.33 Morphological gradient of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Gradient

$$g = (f \oplus b) - (f \ominus b)$$



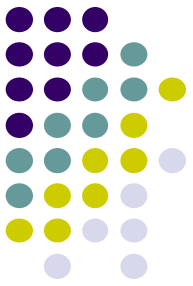
Example 2



a	b
c	d

FIGURE 9.39

(a) 512×512 image of a head CT scan.
(b) Dilation.
(c) Erosion.
(d) Morphological gradient, computed as the difference between (b) and (c).
(Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)



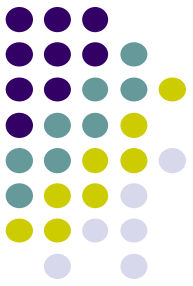
Top-hat and Bottom-hat Transformations

- The top-hat transformation of a grayscale image f is defined as f minus its opening:

$$T_{hat}(f) = f - (f \circ b)$$

- The bottom-hat transformation of a grayscale image f is defined as its closing minus f :

$$B_{hat}(f) = (f \bullet b) - f$$



Example

- Top-hat:
 - Enhancing details in presence of shades

$$h = f - (f \circ b)$$

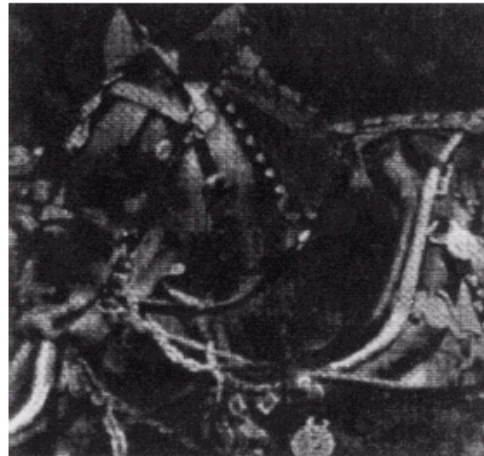
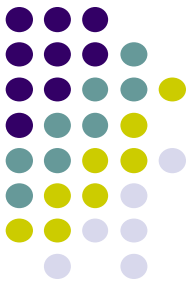


FIGURE 9.34 Result of performing a top-hat transformation on the image of Fig. 9.29(a).
(Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

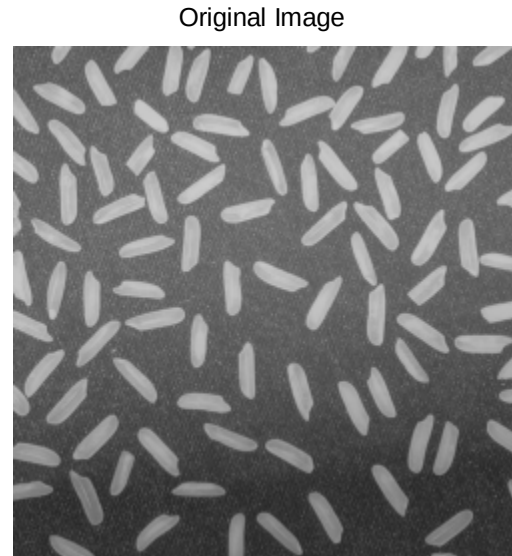


Matlab Code for Top-Hat



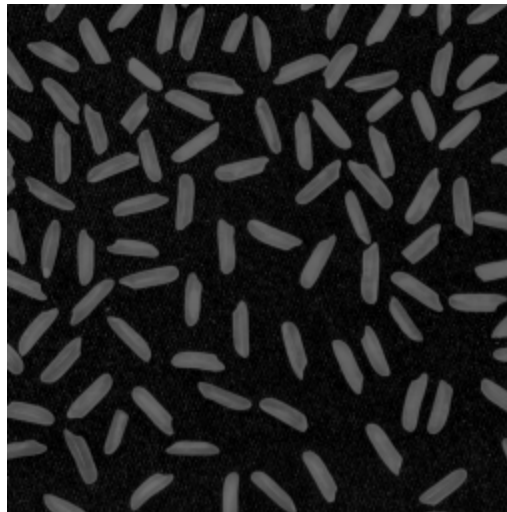
```
im = imread('rice.png');  
imshow(im), title('Original Image ');
```

```
se = strel('disk',12);  
J = imtophat(im,se);  
figure, imshow(J)
```



```
K = imadjust(J);  
figure, imshow(K)
```

Performing TopHat Operation

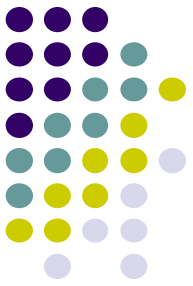


Final Adjusted image



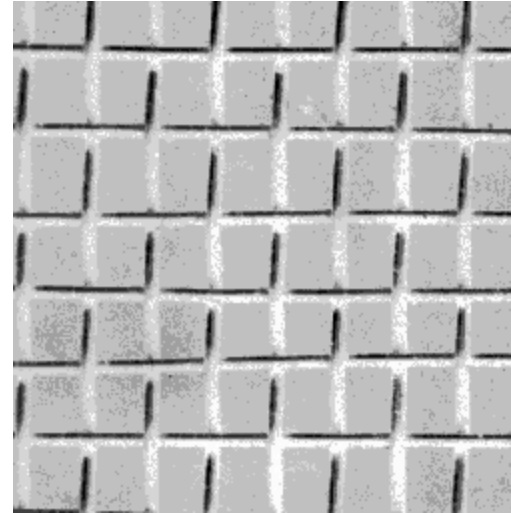


Matlab Code for Bottom-hat

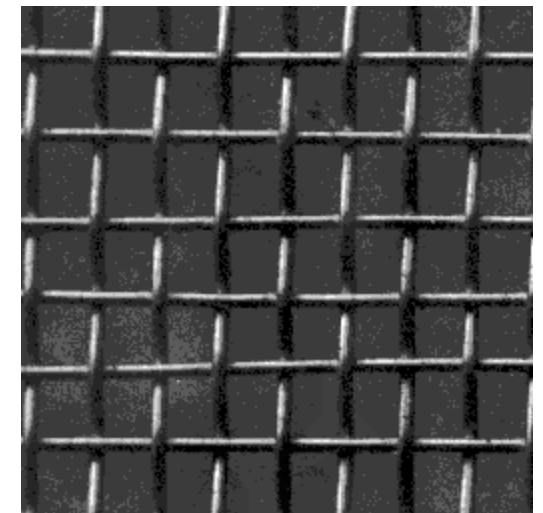


```
im = imread('D1_256_8b.bmp') ;  
  
imshow(im), title('Original Image ');  
  
se = strel('disk',32);  
J = imbothat(im,se);  
figure, imshow(J)
```

Original Image

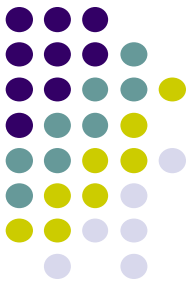


Performing Bottom-Hat Operation





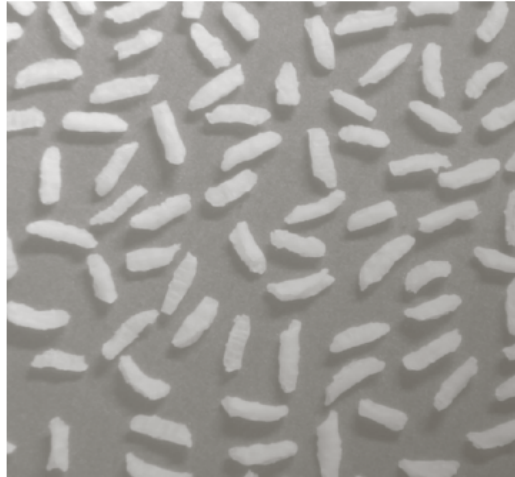
Top-hat and Bottom-hat Transformations



- **One of the principal applications of these transformations is in removing objects from an image by using structuring element in the opening or closing operation**

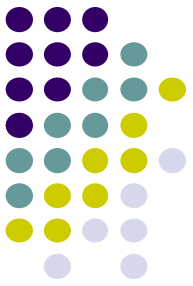


Example of Using Top-hat Transformation in Segmentation



a	b
c	d e

10/3/201: **FIGURE 9.40** Using the top-hat transformation for *shading correction*. (a) Original image of size 320×600 pixels. (b) Thresholded image. (c) Image opened using a disk SE of radius 40. (d) Top-hat transformation (the image minus its opening). (e) Thresholded top-hat image.



Granulometry

- Granulometry deals with determining the size of distribution of particles in an image
- Opening operations of a particular size should have the most effect on regions of the input image that contain particles of similar size
- For each opening, the sum (**surface area**) of the pixel values in the opening is computed



Example 1

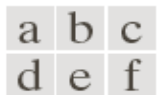
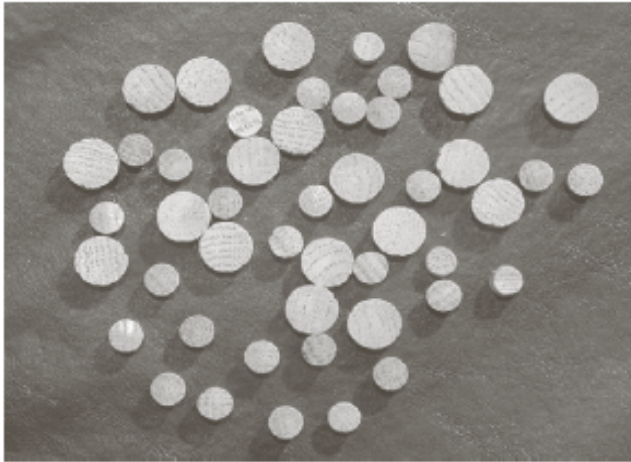


FIGURE 9.41 (a) 531×675 image of wood dowels. (b) Smoothed image. (c)–(f) Openings of (b) with disks of radii equal to 10, 20, 25, and 30 pixels, respectively. (Original image courtesy of Dr. Steve Eddins, The MathWorks, Inc.)



Surface Areas as a function of SE

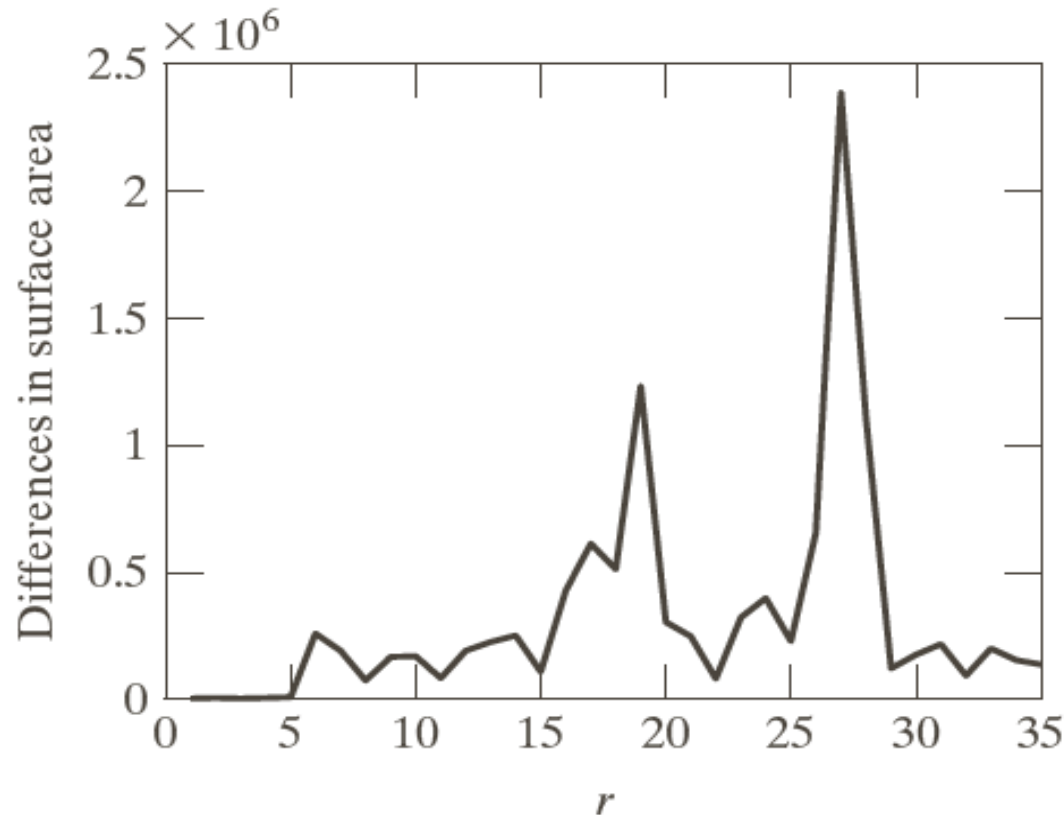
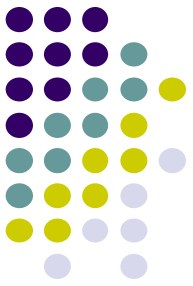
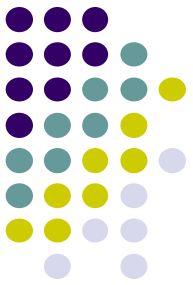
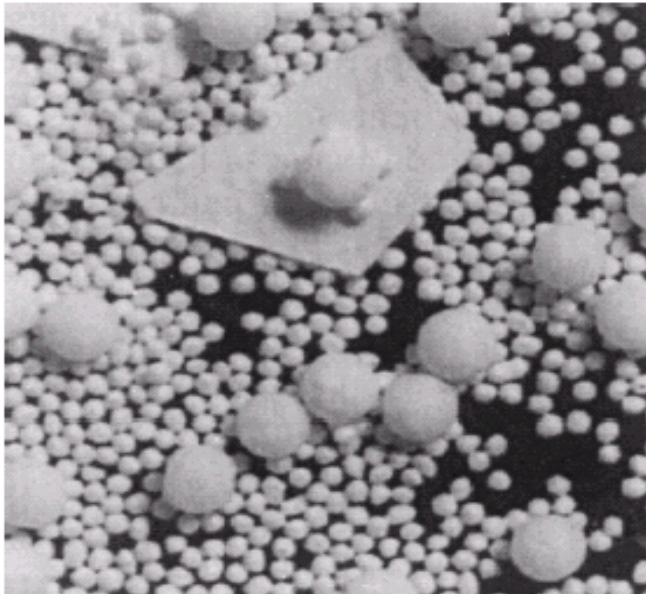


FIGURE 9.42
Differences in surface area as a function of SE disk radius, r . The two peaks are indicative of two dominant particle sizes in the image.



Example 2



a b

FIGURE 9.36

(a) Original image consisting of overlapping particles; (b) size distribution.

(Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

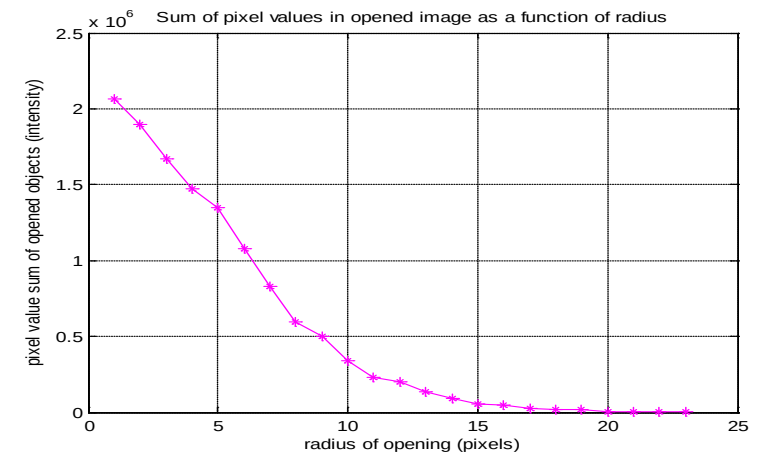
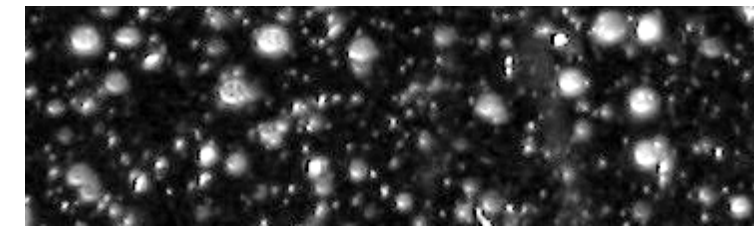
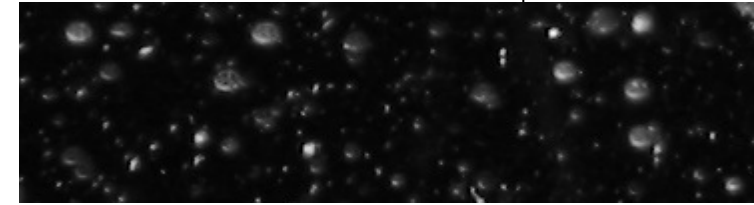


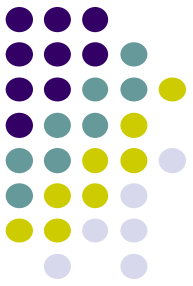
Granulometry Example 3



```
im = imread('snowflakes.png');  
imshow(im), title('Original Image ');  
%% Enhance Contrast  
clahel = adapthisteq(im,'NumTiles',[10 10]);  
clahel = imadjust(clahel);  
figure, imshow(clahel);  
%% Granulometry part:  
for counter = 0:22  
    remain = imopen(clahel, strel('disk', counter));  
    intensity_area(counter + 1) = sum(remain(:));  
end  
figure, plot(intensity_area, 'm - *'), grid on;  
%% To extract certain sizes of objects:  
open5 = imopen(clahel, strel('disk', 5));  
open6 = imopen(clahel, strel('disk', 6));  
rad5 = imsubtract(open5, open6);  
figure, imshow(rad5,[]);
```

Original Image





Textual Segmentation

- **Segmentation:**
the process of subdividing an image into regions.



Example of Textual Segmentation

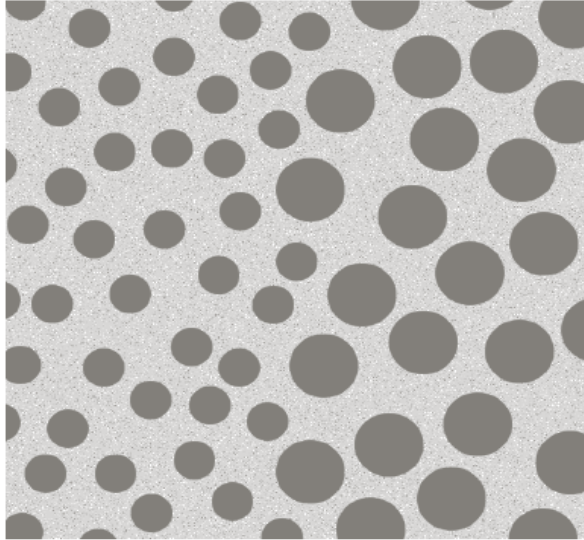


FIGURE 9.43

Textural
segmentation.

(a) A 600×600
image consisting
of two types of
blobs. (b) Image
with small blobs
removed by
closing (a).

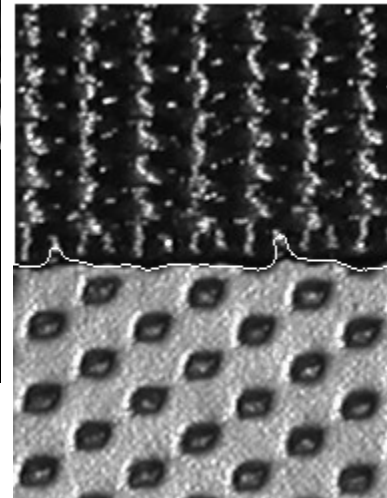
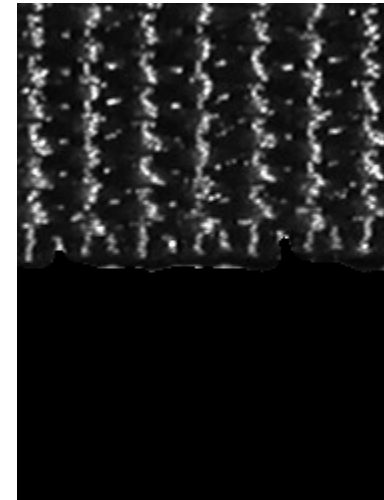
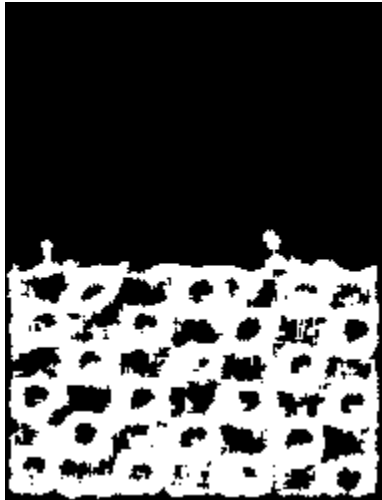
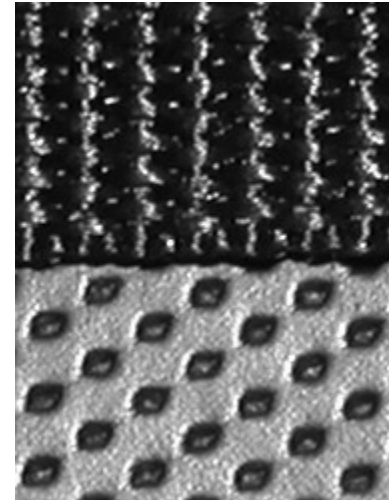
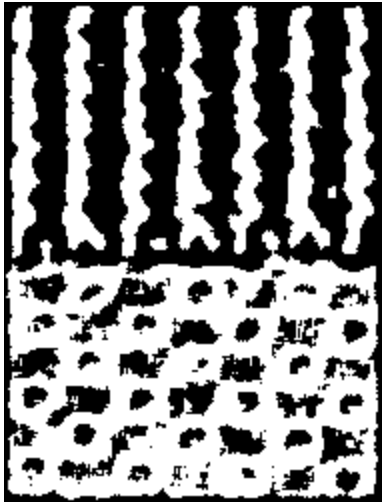
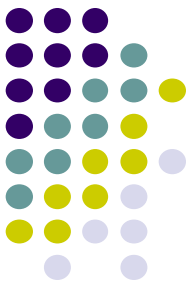
(c) Image with
light patches
between large
blobs removed by
opening (b).

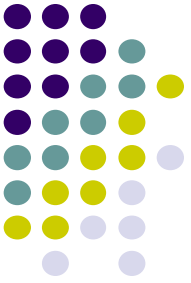
(d) Original
image with
boundary
between the two
regions in (c)
superimposed.

The boundary was
obtained using a
morphological
gradient
operation.



Another Example (Matlab)





Thank you

for your attention